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A Mid-Triassic Paleomagnetic Age of the Kupferschiefer Mineralization in Poland, Based on a Revised Apparent Polar Wander Path for Europe and Russia

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Extensive mineralization occurs in southwest Poland as zoned blankets of Cu-Ag-(Pb-Zn) sulphides in the Kupferschiefer and Zechstein Limestone across the base of the Late Permian Zechstein restricted marine sequence, and in the Weissliegendes sandstone at the top of the Early Permian Rotliegendes continental volcanic and siliciclastic sequence. The Rote Fäule (RF) zone, an oxidized, hematitic equivalent of the normally pyritic basal Zechstein, directly underlies the copper ore and is part of the metal zoning. Although the origin has generally been considered to be syngenetic, geologic controls of the RF-ore systems in Poland suggest a late diagenetic origin by convecting metalliferous brines migrating through the Rotliegendes clastic redbeds up along basement highs, oxidizing the pyritic Kupferschiefer to form the RF, and precipitating base and precious metals on the reduced side of this oxidation-reduction front. A paleomagnetic study of the ore zone in five mines in Poland, the RF zone in three mines, and of the barren pyritic basal Zechstein from outcrops and quarries in Poland, Germany, and England was undertaken to determine the absolute age of the mineralization. In the barren pyritic rocks and the copper sulphide ore, thermal demagnetization and vector subtraction revealed an unstable low-temperature first-remanence component (76.9°N, 12.2°E; A_{63}=1.6°), parallel to the ambient field, and a second, higher temperature component (82.6°N, 145.9°E; A_{63}=1.9°), which could be Miocene (10–20 Ma) or, less likely, Cretaceous (110 Ma) or Jurassic (180 Ma). This second component is erased at 350°–400°C and is thought to be carried by a product of weathering after Alpine age uplift and erosion. No primary magnetization could be isolated in the copper or pyritic zones. The RF hematite, however, contained a very stable third, reversed component (49.0°N, 157.2°E; A_{63}=2.2°) which was revealed typically between 500° and 620°C, and is considered to represent a secondary chemical remanent magnetization acquired during the mineralizing process. In order to date the isolated magnetic components, Carboniferous to Neogene apparent polar wander (APW) paths for Europe and Russia were constructed using paleopole catalogs. Mean paleopoles every 10 Ma from 350 to 0 Ma, with A_{63} circles of confidence, were calculated as running averages using overlapping 20- and 30-Ma time intervals. The RF pole, corresponding to the third remanence component, coincides extremely well with a Middle Triassic age of 240 Ma on the 20-Ma APW path and a Late Triassic age of 230 Ma on the 30-Ma APW path. Statistical analysis yields upper and lower age limits of 250 and 220 Ma. This Triassic paleomagnetic age supports the proposed late diagenetic origin of the Kupferschiefer ore deposits, and suggests that the RF-ore systems were formed immediately after Triassic continental rifting associated with the opening of the Tethys ocean to the south–a likely source of the thermal anomaly necessary to initiate adequate convective velocities within the Rotliegendes clastic basins.

INTRODUCTION

The origin of the Kupferschiefer Cu-Ag deposits at the contact between the Early Permian Rotliegendes continental redbeds and the Late Permian Zechstein restricted marine deposits in central Europe has long been a subject of controversy [*Gregory*, 1930]. Although most workers have concluded that the ore mineralization is syngenetic (see review in *Jung and Knitzschke* [1976]), or with possibly some diagenetic remobilization [*Rydzewski*, 1965, 1978; *Rentzsch*, 1974], geologic evidence suggests that the large Kupferschiefer Cu-Ag deposits in southwest Poland are instead epigenetic [*Lisiakiewicz*, 1969] or late diagenetic [*Jowett et al.*, 1986]. This evidence includes (1) regular metal zoning that crosscuts stratigraphy in section and is highly irregular in plan, (2) replacement of quartz grains and calcite cement by sulphides, (3) replacement of pyrite spherules by hematite and base metal sulphides, and (4) dilatant horizontal and vertical mineralized veinlets that appear to have formed during the main mineralizing event [*Jowett*, 1983].

The ore is directly underlain and controlled by the Rote Fäule (RF), a barren oxidized and reddened portion of the normally reduced pyritic basal Zechstein rocks, which displays irregular spots and streaks of hematite. The RF is an integral part of the metal zoning [*Rydzewski*, 1965, 1978; *Rentzsch*, 1974]. The RF-ore zones form shallow domal shapes and are coincident with a series of underlying buried basement highs and with continental volcanism in the Lower Rotliegendes [*Jowett et al.*, 1986]. These associations and controls suggest that the RF and ore were cogenetic and formed by convecting Rotliegendes formation brines, which carried the base and precious metals leached from the Rotliegendes volcanic detritus through the redbeds and up the flanks of buried basement highs to the reduced pyritic Kupferschiefer above [*Jowett*, 1986].

The Kupferschiefer was deposited in mid-Permian time, and Early Cretaceous faults and basalt dikes crosscut and displace the RF-ore metal zones with little remobilization [*Salski*, 1977]. The timing of the mineralization must then be between these two ages. When the RF was formed, whether contemporaneously with the sediment or later, the hematite should have acquired a chemical remanent magnetization, or CRM [*Stacey and Banerjee*, 1974], aligned with the ambient field at that time. Isolation of this CRM and comparison with the apparent polar wander path for Northern Europe could therefore indicate the absolute age of the RF, and thus of the Cu-Ag mineralization, and help in understanding the origin of these deposits.

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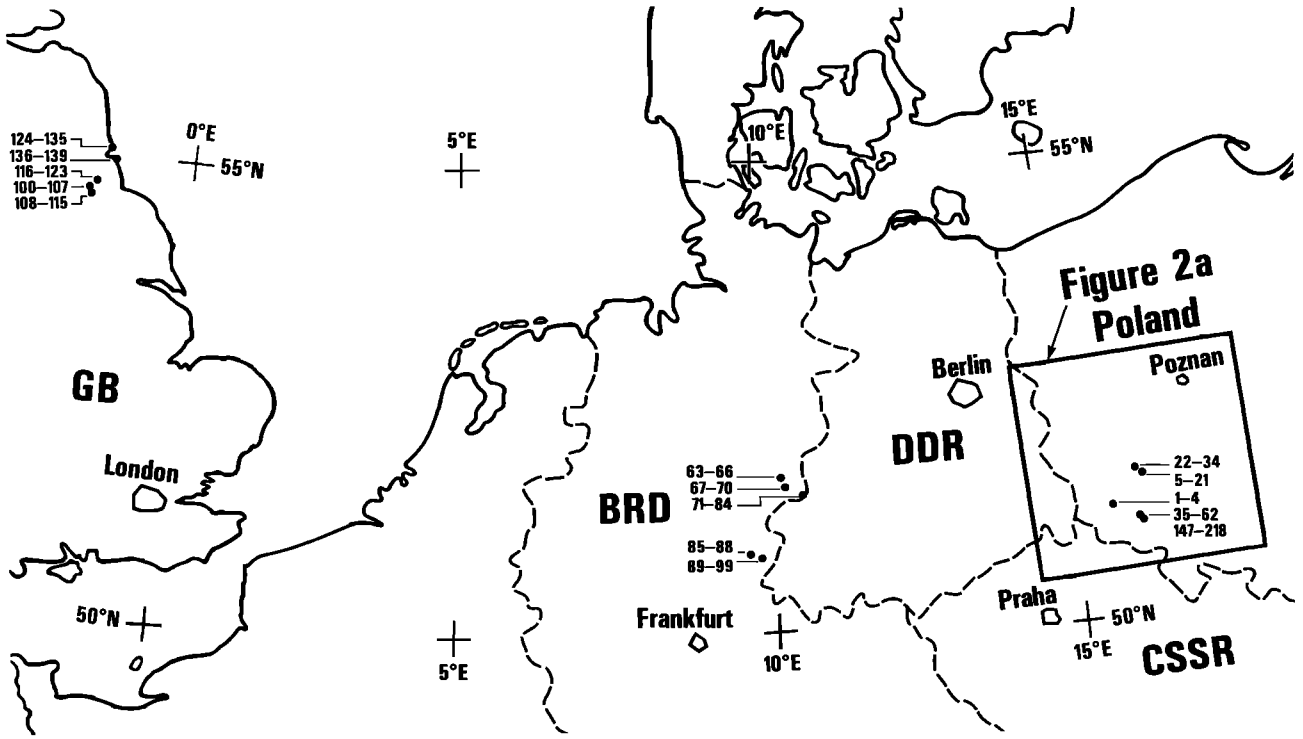


Fig. 1. Location map of outcrops and quarries in England and West Germany where barren pyritic Zechstein rocks were sampled, and of open pit and underground mines in southwest Poland where copper sulphide ore, barren pyritic, and hematitic Rote Fäule samples were obtained.

and the University of Toronto was initiated in 1981 to determine an absolute age of the Rote Fäule and ore zones by paleomagnetic techniques. This paper describes the methods and results of this paleomagnetic study of the timing of the Kupferschiefer mineralization.

MAGNETIC MEASUREMENTS

Oriented hand samples, varying in size from 500 g to 10 kg, were collected from the ore zone underground in the Lubin and Polkowice mines in the Fore-Sudetic Monocline, and from the ore and RF zones in the Konrad (underground), Lena, and Nowy Kosciol (open pit) mines in the North Sudetic Basin in southwest Poland in 1981 and 1983 (Figures 1 and 2). Also, oriented hand samples were taken from outcrops and quarries of barren, pyritic basal Zechstein in West Germany and England (Figure 1) to compare with the Polish samples. The majority of samples were collected in the lower 2 m of the Zechstein, but some were taken as high as 4 m up. Most samples in this study were subhorizontal (carbonates often develop irregular bedding during compaction). Those samples dipping a few degrees were rotated around the strike line to horizontal in the lab and reoriented (i.e., structurally corrected) prior to cutting.

Specimens of rock, of 50 to 500 g mass and irregular in shape (but with upper and lower horizontal planes), were cut from the samples of barren pyritic rock, Rote Fäule, and ore. These were measured on a Develco cryogenic magnetometer and thermally demagnetized in a stepwise manner with a furnace capable of 800°C built by staff of the Paleomagnetic Laboratory at Erindale Campus, University of Toronto. Test samples were also demagnetized by the alternating field technique, but because the hematite in the RF specimens was unaffected, even at 100 milliteslas (mT), it was discontinued. Further measurements of small cylindrical specimens (1-cm diameter; 1–5 g) of RF were made with a Canadian Thin Films Systems cryogenic gradiometer, either with stepwise thermal demagnetization or with continuous thermal demagnetization in a furnace capable of 650°C [*Sugiura et al.*, 1986].

In all, some 474 specimens prepared from 154 samples of ore and barren pyritic facies were measured: 200 specimens each with 4–10 (average 6) measurements (measurement refers to the magnetic vector measured at each demagnetization step); 169 specimens with three measurements; and 105 specimens with only two measurements each. From the RF zones, 114 specimens prepared from 35 samples were measured: 74 specimens with 4 to 14 (average 7) measurements; and 38 specimens with only two measurements each.

The data were reduced by means of the VAX-780 at the Erindale Computing Centre to construct stereographic projections of the measured magnetic moments and of the vectors subtracted between demagnetization steps (Figure 3). Normalized demagnetization curves, Zijderveld plots [*Zijderveld*, 1967], and stability index plots [*Symons and Stupavsky*, 1974] (not shown) were also produced. Line drawings of all graphs and stereographic projections for each specimen were made on an HP-7580B plotter in order to interpret the data visually. Also, all the data were processed with a program described by *Kirschvink* [1980] to determine stable magnetic components.

MAGNETIC COMPONENTS ISOLATED

First and Second Components (Normal) in Ore and Barren Pyritic Rocks

In the barren pyritic facies, mainly from Germany and England (Marl Slate), and in the copper sulphide ore facies from Poland, the natural remanent magnetization (NRM) intensities were typically 8 x 10^{-8} Am^2/kg (Am^2/kg = emu/g), with a few as high as 8 x 10^{-7} to 10^{-6} Am^2/kg. The magnetization was soft to thermal demagnetization, retaining half the NRM intensity at only 200°–250°C, and becoming completely demagnetized at 350°–400°C (Figure 3*a*). The measured moments (MM) typically

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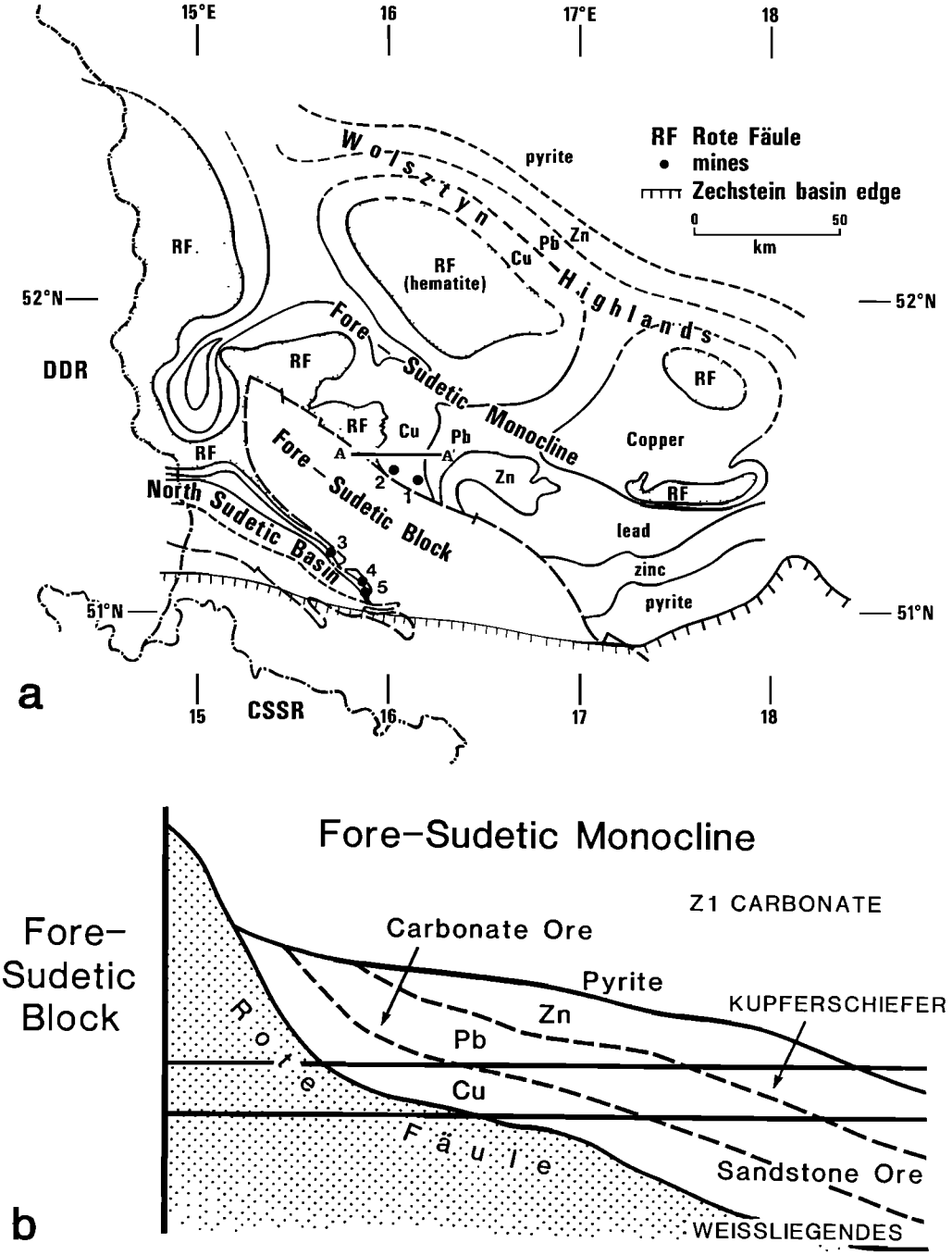
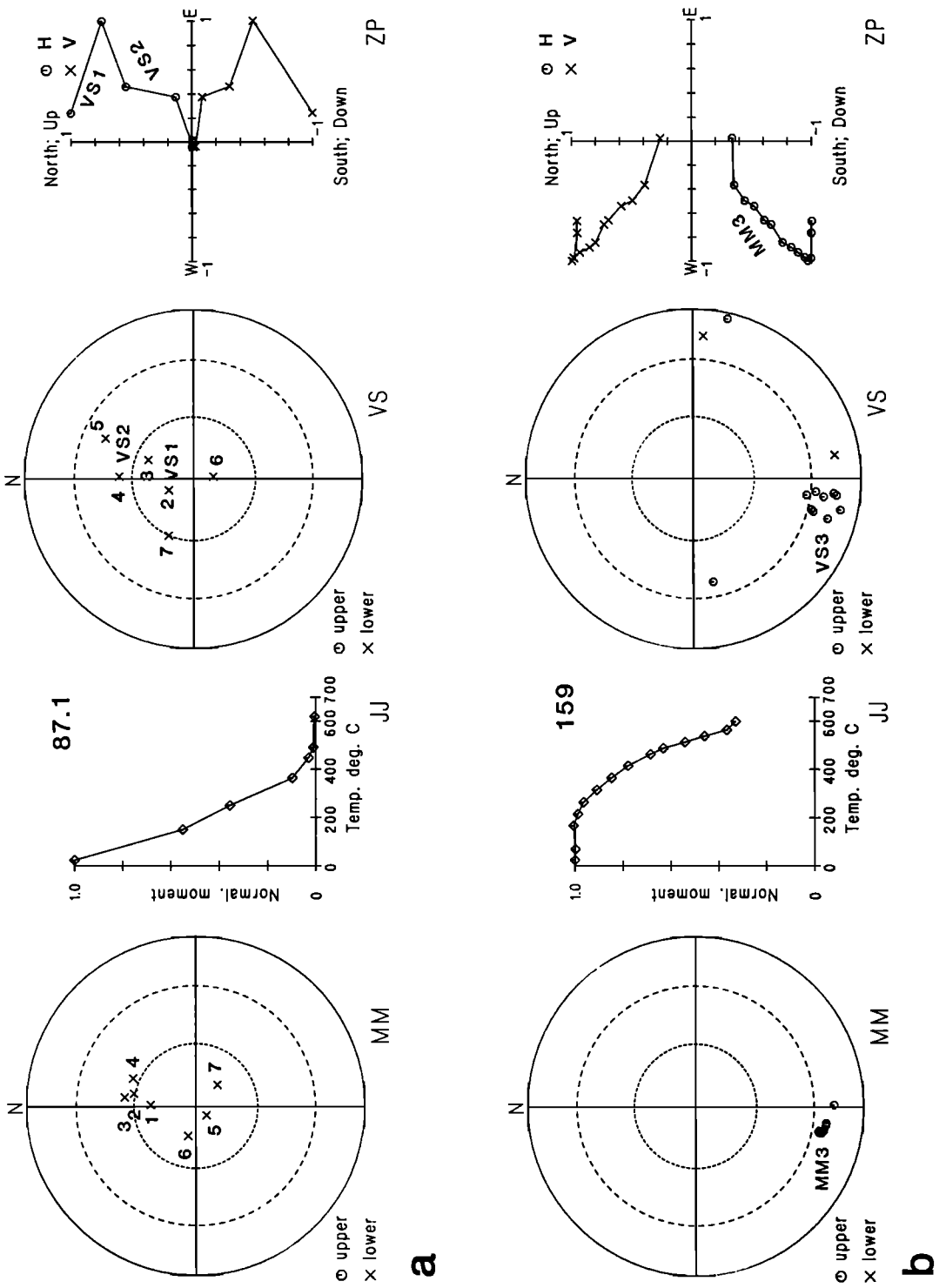


Fig. 2. (*a*) Map of southwestern Poland showing the relationship between the hematitic Rote Fäule (RF), the adjacent base metal ore, and the barren pyritic zones of the basal Zechstein at depths of > 600 m except in the Konrad-Lena area, where it was surface mined. Mines sampled are (1) Lubin, (2) Polkowice, (3) Konrad, (4) Nowy Kosciol, and (5) Lena. Metal zoning after *Rydzewski* [1978]. Collection sites from outcrops and quarries in West Germany and England are located in Figure 1. (*b*) Schematic section across the Lubin mine area (A-A' in Figure 2*a*) depicting the vertical metal zonation and the integral nature of the RF in this zoning. The RF-ore zone cuts across the stratigraphy from the Z1 Carbonate and Kupferschiefer of the basal Zechstein restricted marine sequence, down into the Weissliegendes sandstone at the top of the Rotliegendes continental volcanic and siliciclastic sequence. The Fore-Sudetic Block is a tectonic uplift resulting from Alpine age movements.

moved from a steep normal NRM direction to a shallower inclination, sometimes clustering until demagnetized. The vectors subtracted (VS), which can help isolate components in multicomponent rocks more so than the MM [*Hoffman and Day*, 1978], were separated into two groups, VS1 and VS2. VS1 is the first component subtracted between the NRM and the first demagnetization step (usually 100°–200°C), and the second component VS2, is the vector resultant of typically two to five directions in each specimen that appeared to form a rough clustering. Of the original 474 specimens, VS1 was calculated from 373 specimens (126 samples), and VS2 was calculated using 247 specimens (85 samples).

VS1 was consistently steeper in inclination than VS2 (e.g., average directions for Polish rocks were $D = 3.3°$, $I = +74.1°$ and

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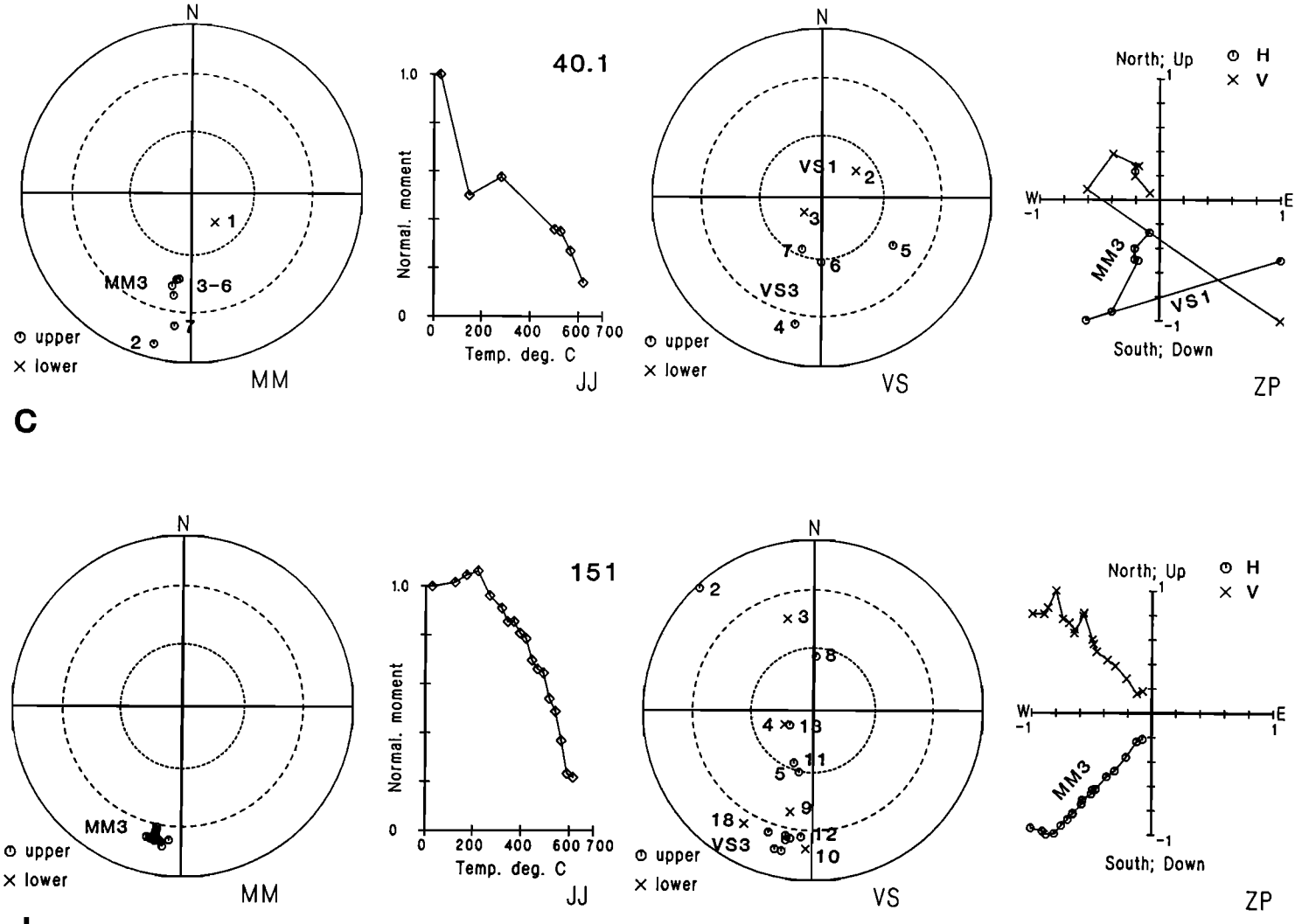


Fig. 3. Graphical representations of the thermal demagnetizations of (*a*) specimen 87.1 of pyritic rock containing the low-temperature first (VS1) and second (VS2) remanence components. VS2 in this specimen is the average of the third, fourth, and fifth vectors subtracted. (*b*) Red hematitic Rote Fäule specimen 159 with resistant magnetization, and containing only the third, reversed component, reflected in the demagnetization plot which displays measured moments (MM3), and the vector difference plot which displays vectors subtracted between each demagnetization step (VS3). (*c*) Mottled red and black Rote Fäule specimen 40.1 with both low temperature magnetization (VS1, possibly VS2) and resistant magnetization (VS3 carried by hematite) (*d*) Red hematitic Rote Fäule specimen 151 with stable MM reflecting the third component MM3, but with the VS directions making northerly "shifts". These shifts may represent magnetic noise during measurement caused by the dehydration of goethite and associated formation of hematite. MM, measured moment plots on equal-area projection; JJ, normalized intensity curves; VS, vector subtracted between measurements; and ZP, Zijderveld plots.

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TABLE 1. The First and Second Remanence Components in the Sulphide-Bearing Basal Zechstein in Europe

	Component Isolated						Paleopole Position					
	Declination, deg	Inclination, deg	N	k	α_{63}, deg	α_{95}, deg	Latitude, °N	Longitude, °E	N	k	A_{63}, deg	A_{95}, deg
First Vector Subtracted (VS1)												
Poland	3.3	74.1	51	51	1.6	2.8	79.7	27.7	51	19	2.6	4.5
Germany	0.6	75.1	39	58	1.7	2.9	78.2	10.3	39	23	2.7	4.7
England	4.1	81.6	36	94	1.4	2.4	70.7	1.8	36	28	2.6	4.4
VS1 Total Europe							76.9	12.2	126	21	1.6	2.7
First Kirschvink Component (KV1)												
Poland	11.7	75.1	34	76	1.6	2.8	77.0	41.4	34	25	2.8	4.8
Germany	5.5	71.7	22	56	2.3	4.0	83.1	32.4	22	23	3.6	6.2
England	12.4	78.6	34	51	2.0	3.4	74.3	15.4	34	18	3.3	5.6
KV1 Total Europe							77.8	28.0	90	21	1.9	3.2
Second Vector Subtracted (VS2)												
Poland	7.8	60.8	28	40	2.4	4.2	80.2	162.3	28	23	3.2	5.5
Germany	6.8	66.8	24	53	2.3	3.9	86.0	109.3	24	29	3.1	5.3
England	6.5	64.0	33	31	2.5	4.4	82.3	141.9	33	17	3.4	5.9
VS2 Total Europe							82.6	145.9	85	21	1.9	3.3
Second Kirschvink Component (KV2)												
Poland	8.2	59.3	26	28	3.0	5.1	78.6	163.4	26	17	3.9	6.7
Germany	2.5	65.1	23	59	2.2	3.8	86.2	166.0	23	30	3.1	5.4
England	9.9	64.1	21	38	2.9	5.0	79.7	130.1	21	21	3.8	6.6
KV2 Total Europe							81.8	151.9	70	21	2.1	3.7

N is the number of samples used in the calculation; k is the *Fisher* [1953] precision parameter; α_{63}, α_{95}, A_{63}, and A_{95} represent the radii of error circles (determined from remanence directions (α) or virtual geomagnetic poles (A)), within which there is a 63% or 95% chance that the true direction of the population lies.

D = 7.8°, I = +60.8°, respectively). The VS2 component was the stable end component present in the basal Zechstein rocks in Germany and England. The summary in Table 1 (see also Figure 4*a*) shows that VS1 and VS2 are the same whether from England, Germany, or Poland.

The lowest temperature components isolated by *Kirschvink*'s [1980] program (KV1) in the three regions (using a 5° standard deviation angle for line segments) are similar to the respective VS1 components, and the highest temperature components (KV2) are similar to the respective VS2 components (Table 1 and Figures 4*a* and 4*b*). This similarity demonstrates that the vector subtraction method is comparable to the method recommended by *Kirschvink* [1980].

Those specimens not used to calculate VS1 (21% not used) typically had only two to three measurements, and the temperature spread between the first two measurements was too great to be representative of VS1. The specimens not used for VS2 (47%) either had too few measurements or did not develop a clustering of directions at higher temperatures. Those not used for KV1 and KV2 simply did not meet the 5° angle criterion in *Kirschvink*'s [1980] program. The VS1, VS2, KV1, and KV2 components were plotted both as points and as contoured Gaussian density diagrams [*Robin and Jowett*, 1986] for each geographic region (example in Figure 5). Points lying outside the main cluster were also discarded.

The combined paleopoles for Europe (Figure 4) show that the first, low-temperature component (VS1 and KV1) lies between Europe and the north pole, and the second, higher temperature component (VS2 and KV2) coincides closely with the geographic north pole.

Third Component (Reversed) in Rote Fäule

In the oxidized Rote Fäule facies from the Konrad, Lena, and Nowy Kosciol mines in Poland, NRM intensities were typically 5 x 10^{-7} to 2 x 10^{-6} Am^2/kg, but as high as 8 x 10^{-6} Am^2/kg in purely red specimens. The magnetization was very resistant to thermal demagnetization, retaining half the NRM intensity in the red specimens at 400°–550°C, and becoming unmeasurable only well above 600°C. In the purely red specimens, only a negative component was generally present (Figure 3*b*), reflected by the negative inclinations of the MM and VS plots with southerly declinations, and the square-shouldered demagnetization curves, typical for hematite with a Curie temperature (T*c*) of 680°C [*Stacey and Banerjee*, 1974].

In specimens mottled red and black, a normal magnetization similar to VS1 and VS2 components was erased before the reversed end point, stable from 300°C to above 600°C, was reached (Figure 3*c*). This is reflected in the demagnetization curves in which a significant proportion of the NRM is lost by 200°C, with a square-shouldered curve at higher temperatures. These curves are typical for rocks containing hematite, a high T*c* carrier, combined with goethite or maghemite, which may retain their magnetization up to 250°–350°C [*Strangway et al.*, 1968, 1969; *Hedley*, 1968; *Stacey and Banerjee*, 1974], or monoclinic pyrrhotite with a T*c* of 320°C [*Schwarz*, 1975]. In other mottled samples, especially from the Konrad mine, the normal component was very resistant and was erased only at temperatures of 500°C, which suggests that it could be carried by hematite or perhaps magnetite (T*c* of 580°C).

Many of the copper sulphide ore specimens, usually those from

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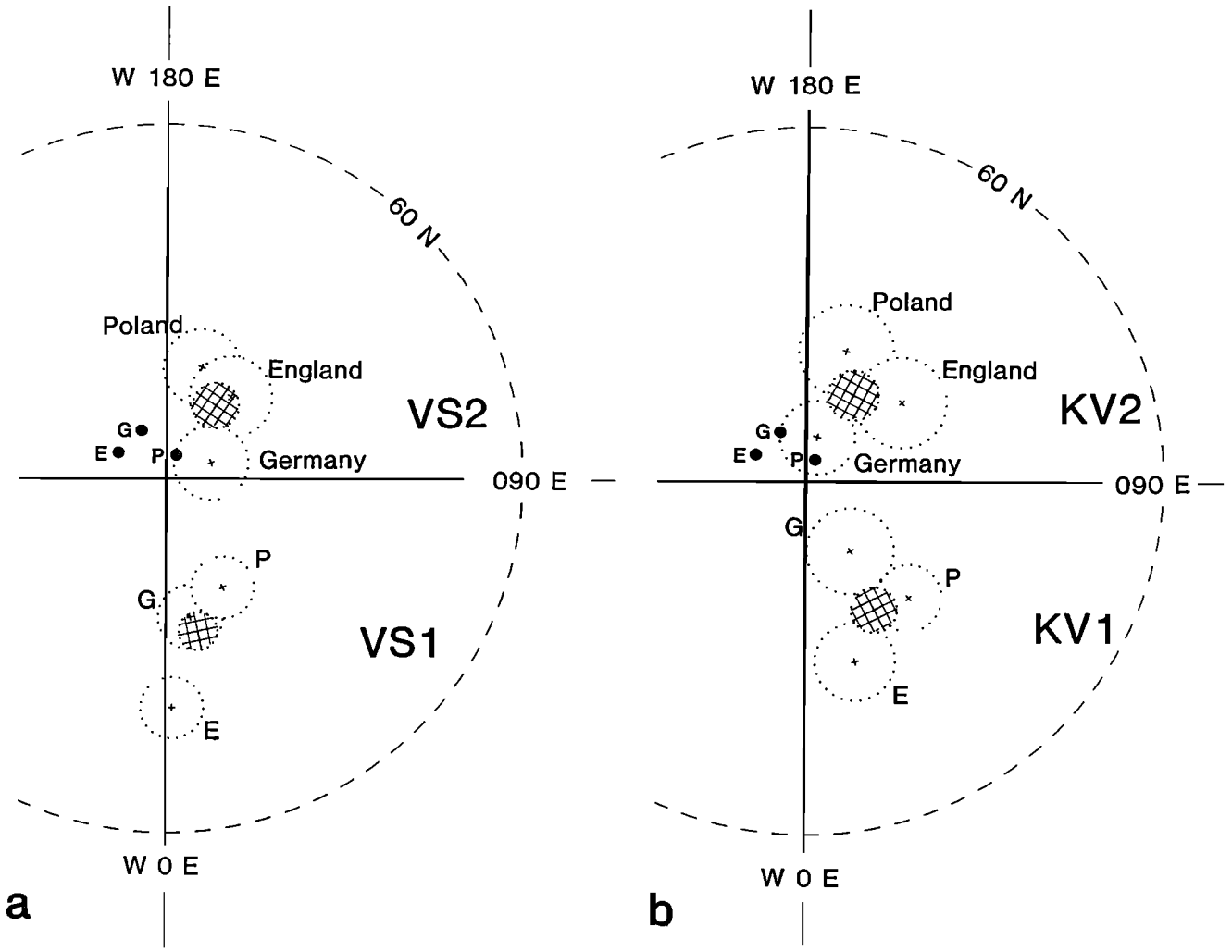


Fig. 4. Comparison of results from the (*a*) vector subtracted (VS) and (*b*) *Kirschvink* [1980] (KV) methods for calculating the first and second magnetic components represented as north VGP's with A_{63} circles, plotted on north polar, equal-area projections. The McFadden-Lowes statistical test showed that the VS1 and KV1 Europe directions (hatched pattern) are the same, and the VS2 and KV2 Europe directions (hatched pattern) are the same at the 95% confidence level (see text), indicating that either method gives the same results. Present dipole field at each geographic region is represented by solid circles.

the very base of the Kupferschiefer in the Lubin and Polkowice mines, trended toward a reversed direction. The VS generally indicated only a normal component being destroyed, but the MM moved toward a reversed direction (which it could not attain before acquiring a highly viscous magnetization, typically at 500°–550°C). This trend toward a reversed direction was not apparent in the barren pyritic rocks in Germany and England, suggesting that the reversed direction is connected with the mineralizing event.

One interesting result discovered when isolating the third components in the RF samples, especially from the Lena mine, was the presence of anomalous "shifts" of the VS directions toward the northern hemisphere (Figure 3*d*). The MM were normally tightly grouped in the reversed direction at all temperatures and did not move with the VS. Because these normal polarity VS were interspersed with reversed polarity VS and occurred at temperatures as high as 500°–550°C, they cannot represent the lower temperature VS2 component. These samples were characterized by a change in slope on demagnetization diagrams, typically between 300° and 350°C (Figure 6), which is the temperature range in which goethite and maghemite rapidly lose their magnetization [*Strangway et al.*, 1968; *Stacey and Banerjee*, 1974]. Therefore these VS shifts possibly represent magnetic noise caused by the dehydration of iron oxides and the associated formation of hematite.

Calculation of the Third (Reversed) Component

The reversed directions of those RF specimens that reached a stable end point (not just tended toward one) were used to determine the third, reversed paleopole, calculated as MM3, VS3, MMVS3, and KV3 directions (Table 2). The MM3 component was calculated using measured moments (MM) from the Konrad, Nowy Kosciol, and Lena mines; the VS3 component using vectors subtracted (VS) from the Konrad and Nowy Kosciol mines; the MMVS3 component using a combination of MM from Lena and VS from Konrad and Nowy Kosciol mines; and the KV3 component from all mines using *Kirschvink*'s [1980] program. The MM3 and MMVS3 methods were attempted because many VS3 directions were grossly affected by magnetic noise and shifts, and because the Kirschvink method was not successful in determining magnetic components in many RF specimens containing stable end points.

Although measured moments typically do not reflect the lower temperature remanence components being destroyed upon demagnetization (e.g., VS1 and VS2), they can reflect a stable end

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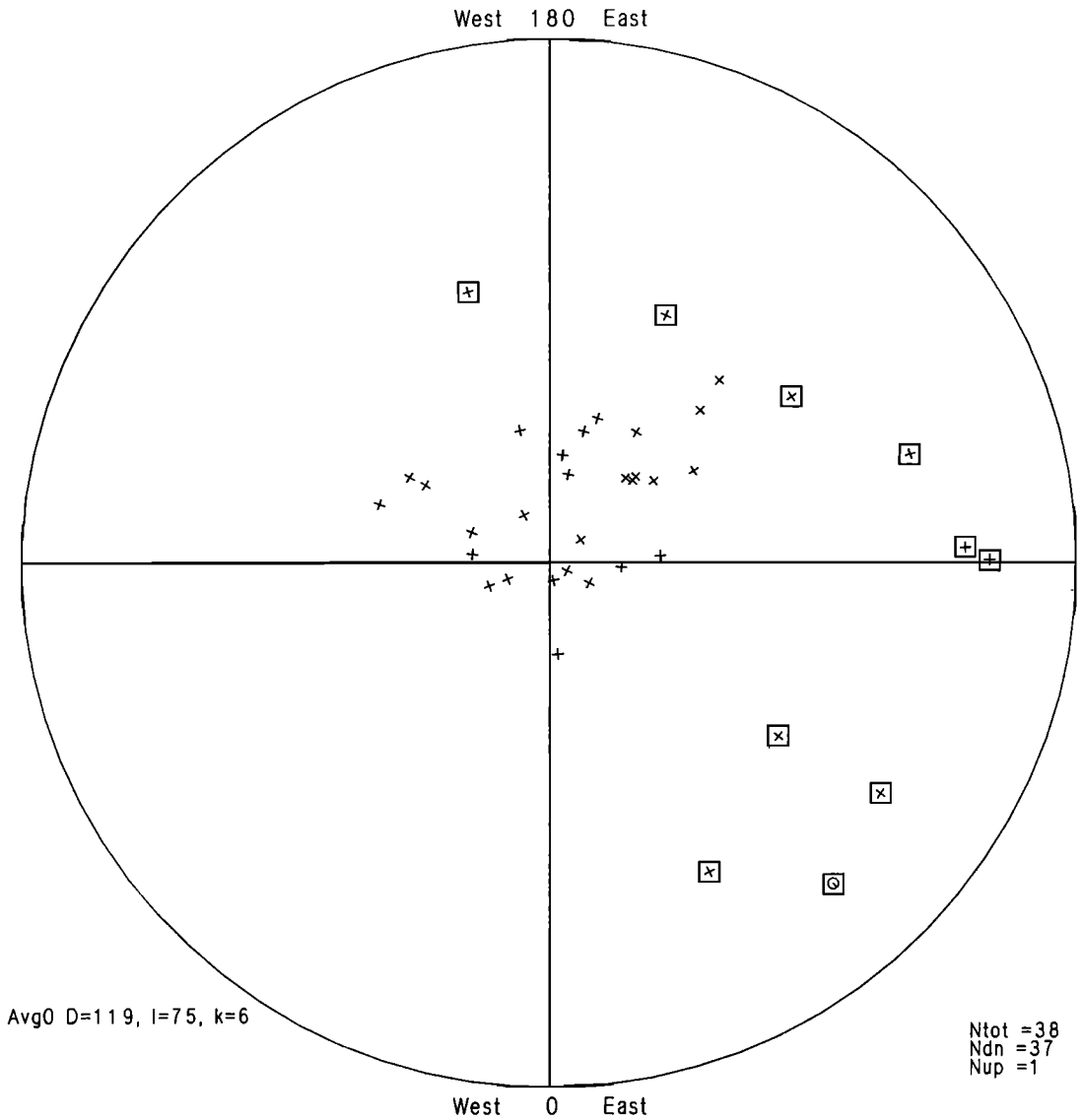


Fig. 5a

Fig. 5. Example of graphical representation of north virtual geomagnetic poles as (*a*) point plots (crosses, lower; circles, upper) and (*b*) contoured Gaussian density diagrams on north polar, equal-area projections. This example shows the VS2 Poland component isolated in 38 samples (103 specimens), of which 10 samples (in boxes) outside of the main cluster were rejected. NO is the number of points (samples) on the hemisphere, E is the expected density if the points were uniformly distributed over the half-sphere, k is the precision parameter used in the Gaussian counting function, and σ is the standard deviation [see *Robin and Jowett*, 1986]).

component as well as the vector subtraction method can, and the MM are not as subject to magnetic noise as the VS3 are. The measured moment (MM3), with a direction of D = 205.4°, I = -28.2° and a paleopole of 49.0°N, 157.2°E; A_{63}=2.2°, is considered the best estimate (smallest A_{63}) of the stable end component in the Rote Fäule. Of the original 114 specimens of RF, only 70 specimens (29 samples) were used to calculate MM3 (Figure 7). Specimens that were mottled or had few hematite spots often tended toward, but did not reach, an end point.

STATISTICAL DISCRIMINATION BETWEEN DIRECTIONS

It would be helpful to know if each of the three components is statistically different from the others, and whether the various methods used (VS, KV, MM) give similar results for each component. *McFadden and Lowes* [1981] have devised an exact test for comparing directions when the precision parameters (k; *Fisher* [1953]) are equal ($k1$=$k2$), and an approximate test when the k's are unequal. The statistic f, derived by McFadden and Lowes, is an approximation of the sampling distribution of the angle γ between the two mean sample directions being compared and is conditional on the observed lengths of the vector resultants. (The assumption is made that the two samples were drawn randomly from *Fisher* [1953] populations sharing a common true mean direction (the null hypothesis).) To test this null hypothesis, the statistic f is compared with the statistic f_p which represents the value of f that will be exceeded with a given probability p. The null hypothesis is rejected if $f > f_p$, or accepted if not. Rejection of the hypothesis at p = 0.05 (95% level of confidence) means that an angle as large as the observed angle γ can be expected in fewer than 5% of the samples taken randomly from two populations with a shared common mean direction.

First component. McFadden and Lowe's test demonstrates

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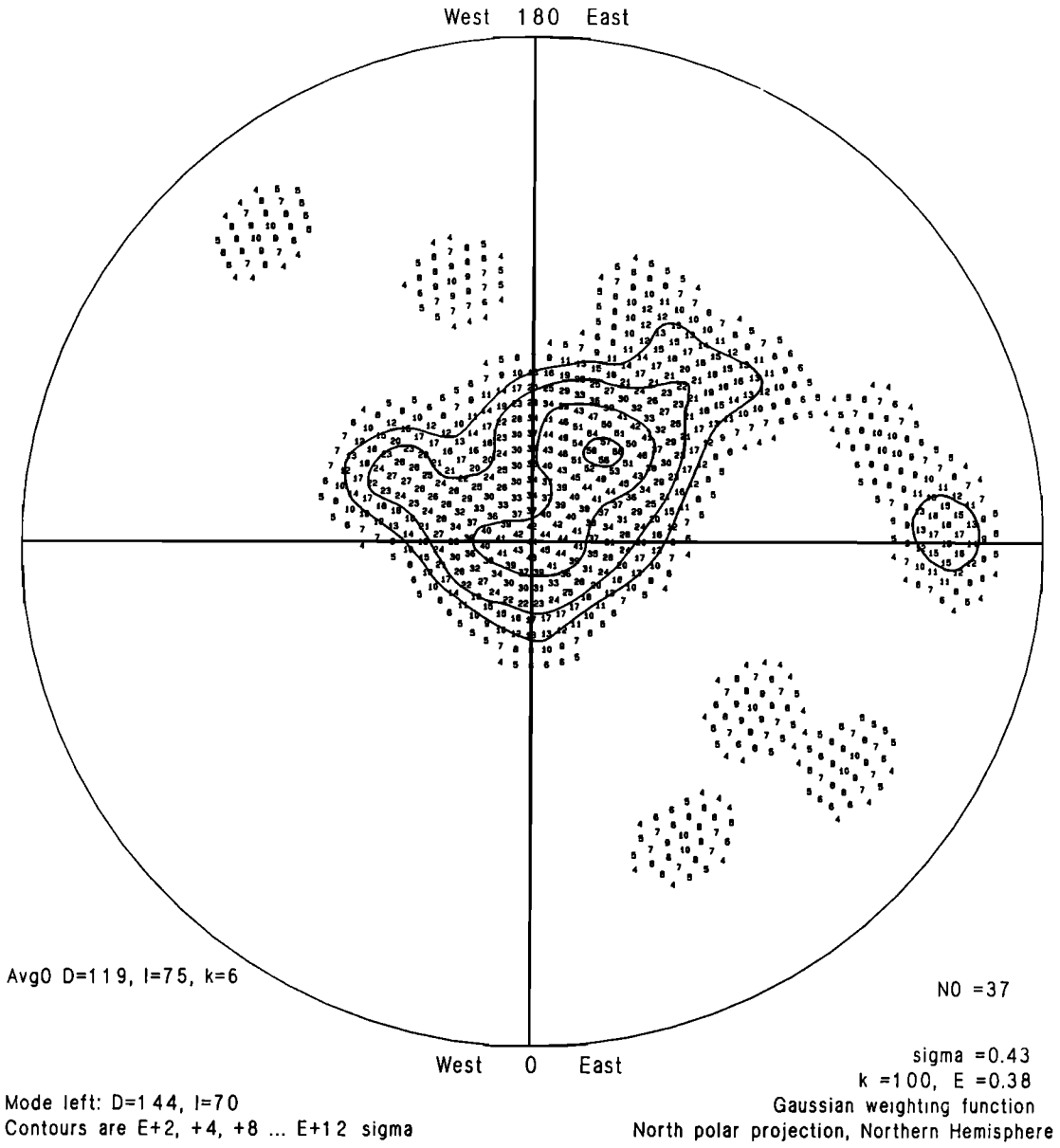


Fig. 5*b*

that VS1 Europe (76.9°N, 12.2°E; k=21) and KV1 Europe (77.8°N, 28.0°E; k=21) are the same for the usual confidence levels of 95% or 99%.

Second component. Likewise, the VS2 Europe (82.6°N, 145.9°E; k=21) and KV2 Europe (81.8°N, 151.9°E; k=21) components are the same. These tests show that the components are similar whether selected by the more subjective vector subtraction method or the more objective Kirschvink method.

Although the KV method gives the same results, the vector subtracted directions (VS1 and VS2) are chosen as best representatives of the first and second components because they have the larger data sets and thus smaller A_{63} (see Table 1). The A_{63} circle, equivalent to the standard error of the mean (p = 0.37) [*Merrill and McElhinny*, 1983], is the area on the sphere within which there is a 63% chance that the true direction of the population lies. *Morel and Irving* [1981] recommend the use of this standard error because, as a rough approximation, if the circles overlap, the directions are not likely to be statistically different, whereas with the more commonly used A_{95} (p = 0.05), the circles may overlap even if the paleopoles are significantly different at p = 0.05 [*Watson and Irving*, 1957].

Third component. Comparison between the MM3 Poland (49.0°N, 157.2°E; k=46) component and each of the other estimates of the third component—VS3 Poland (50.6°N, 159.1°E; k=33), MMVS Poland (48.3°N, 154.7°E; k=32), and KV3 Poland (47.6°N, 156.5°E; k=28))—shows that the null hypothesis (each pair of directions shares a common mean true direction) cannot be rejected at any level of confidence from 99% to 50%. In other words, the third-component directions are all extremely similar and indistinguishable regardless of the method of determination. The measured moment (MM3) direction is chosen as best representative of the third component because (1) it has the smallest standard error A_{63}, (2) because it corresponds more closely to the mean direction of all the methods (Table 2), and (3) because this method can be used for all three mines sampled for RF.

First, second, and third components. Comparison between each of the three isolated components (VS1 Europe, VS2 Europe, and MM3 Poland) shows that the null hypothesis, that they share

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TABLE 2. The Third, Reversed Remanence Component in the Hematitic Rote Fäule in Poland

Component Isolated						Paleopole Position					
Declination, deg	Inclination, deg	N	k	α_{63}, deg	α_{95}, deg	Latitude, °N	Longitude, °E	N	k	A_{63}, deg	A_{95}, deg
Third Moment Measured (MM3)											
205.4	-28.2	29	37	2.5	4.2	49.0	157.2	29	46	2.2	3.9
Third Vector Subtracted (VS3)											
203.5	-29.9	20	32	3.2	5.6	50.6	159.1	20	33	3.2	5.5
Third Subtracted and Measured (MMVS3)											
207.3	-28.5	30	31	2.7	4.6	48.3	154.7	30	32	2.6	4.5
Third Kirschvink Component (KV3)											
206.4	-25.6	17	21	4.2	7.3	47.7	156.6	17	28	3.7	6.4

N, k, α_{63}, α_{95}, A_{63}, and A_{95} as in Table 1.

a common true mean direction, can be rejected at any level between 99% and 50% confidence. In other words, each of the three magnetic components isolated is statistically distinct.

EUROPE AND RUSSIA APPARENT POLAR WANDER PATH

Continental Reconstruction

In order to determine accurately the absolute ages of the isolated magnetic components, an apparent polar wander (APW) path was constructed with segments or subpopulations of known age. Because the paleopole data base for the late Paleozoic and Mesozoic is limited for Europe, the plate tectonic reconstructions of *Scotese et al.* [1979] for Europe and Asia, based on geologic and paleomagnetic information, were used to increase the number of poles available for the APW path.

It is assumed from *Scotese et al.* [1979] that Poland was contained within the stable continent, Baltica, which included Scandinavia and European USSR ("Russia") since the Cambrian; Scotland since the Early Devonian; England and Wales since the Late Carboniferous; and southern Russia and central Europe south of the Hercynian fold belt (France, Germany, and Czechoslovakia) since the Early Permian. Any poles in these time periods from these locations therefore would be representative of Poland as long as there were no large-scale intracratonic plate rotations (with an axis within or near Baltica) and as long as poles from the Alps, Pyrenees, Caucasus, or Urals were not used. *Morel and Irving* [1981] constructed a path from latest Carboniferous to Mid-Jurassic using 73 paleopoles from Europe and northern Asia that met their criteria. However, many of their poles were from Siberia (e.g., 11 of 22 Triassic poles), which have pole longitudes consistently westward of European Russia pole longitudes. While this was necessary for their purpose of continental reconstructions, Siberia is so distant from Poland that the APW path representative of Poland could easily be distorted by plate rotation, and thus Siberian poles were not included in the APW path construction.

Construction of the Path

An apparent polar wander path from 350 to 0 Ma was therefore constructed using data from the paleomagnetic pole catalogs of *Hicken et al.* [1972] and *Irving et al.* [1976*a*, *b*, *c*]. All poles with the A**, A*, and A ratings of *Irving et al.* [1976*a*, *b*, *c*] were included, and B-rated poles were used only when they agreed with the A-rated poles. Poles that lay outside the main density grouping were considered anomalous and discarded (see Figure 5 for an example of this discarding method). The assigned ages were changed from the time scale of *Harland et al.* [1964] to that of *Harland et al.* [1982], using the stratigraphic information supplied in the catalog notes. Radiometric ages were used as is. The poles used in the construction are listed in Table 3.

A total of 340 useable paleopoles were included, but they are unevenly distributed in time, with a high concentration between 0 and 60 Ma and between 230 and 310 Ma, but with fewer than 10 paleopoles for each 30-Ma interval between 100 and 220 Ma. The poles of each geologic period were presented as point plots on north polar equal-area projections and as contoured density diagrams using a Gaussian weighting function [*Robin and Jowett*, 1986], with each pole given a subjective weighting of 3, 2, or 1 (depending on its rating of A**, A*, or A, B) in an attempt to better estimate the real distribution.

Most geologic periods were represented by fairly tight Gaussian distributions of poles with well-defined modes. Also, because the means calculated separately for Europe and Russia from the Carboniferous to Cretaceous were within 2° to 3° of each other, except for the Jurassic (about 8°), the reconstructions of *Scotese et al.* [1979] and the assumption of no major plate rotations between Europe and Russia were accepted as reliable. Poles from the Jurassic period formed a poorly defined elongated cluster, reflecting in part the lack of suitable Jurassic volcanic rocks and redbeds in the undisturbed regions of Europe and Russia, and in part the apparent rapid polar movement during this time.

The paleopoles were then divided into overlapping 20- and 30-Ma time intervals centered every 10 Ma from 350 to 0 Ma, in a manner similar to that of *Morel and Irving* [1981], with the means and the *Fisher* [1953] statistics calculated for each time interval (Table 4). The paths were calculated in two ways: (1) giving each paleopole unit weight and (2) weighting each pole either 3, 2, or 1 according to its reliability [*Irving et al.*, 1976*a*, *b*, *c*]. The weighting in each time interval population was normalized to unity so that the equivalent number of poles (N) in each method was the same and the two could be compared statistically.

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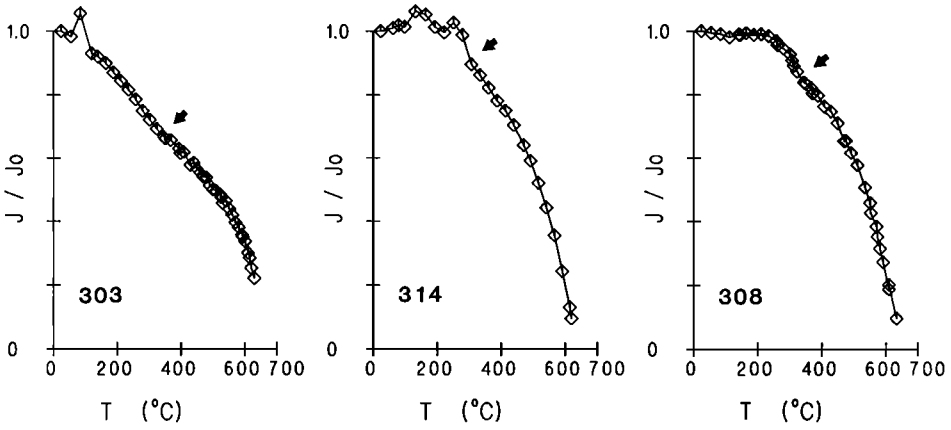


Fig. 6. Normalized demagnetization curves of Rote Fäule specimens (303, 314, 308) showing the dip in intensity (arrows) between 300° and 350°C, which may represent the rapid loss of magnetization as goethite is dehydrated, leaving only hematite as the carrier.

This weighting increased the clustering slightly in the Carboniferous, Early Jurassic, Late Cretaceous, and mid-Tertiary, but decreased it in the Permian, Triassic, and Early Cretaceous, with no effect in the remainder. The overall effect of weighting was one of decreasing the clustering (mean k values were slightly less in the weighted paths), indicating that, given an adequate sampling, the A**, A*, A, and selected B paleopoles from the catalogs of *Irving et al.* [1976*a*, *b*, *c*] can be treated as being equally reliable. Since the unit-weighted APW paths are insignificantly different from the subjectively weighted path (each A_{63} circle of confidence overlaps its age equivalent), the unit-weighted versions will be considered a better estimate of the true path, and all further reference will be to these paths only.

DESCRIPTION OF THE APW PATHS

The 20-Ma Interval Path

The unit-weighted 20-Ma APW path from 350 to 150 Ma (Table 4), with A_{63} circles, describes a fairly detailed curve, with paleopole latitudes ranging between 31°N and 78°N. The detail of the 20-Ma path allows several aberrations or noteworthy events to be seen. First, there is a loop during the Late Carboniferous, when the rapid southeasterly movement from 41°N to 31°N between 340 and 320 Ma reverses to a rapid northward movement to 39°N between 320 and 290 Ma. (When the angles between the end points of the loops are greater than the sum of the A_{63} radii, the loops represent statistically significant movements: accurate dif-

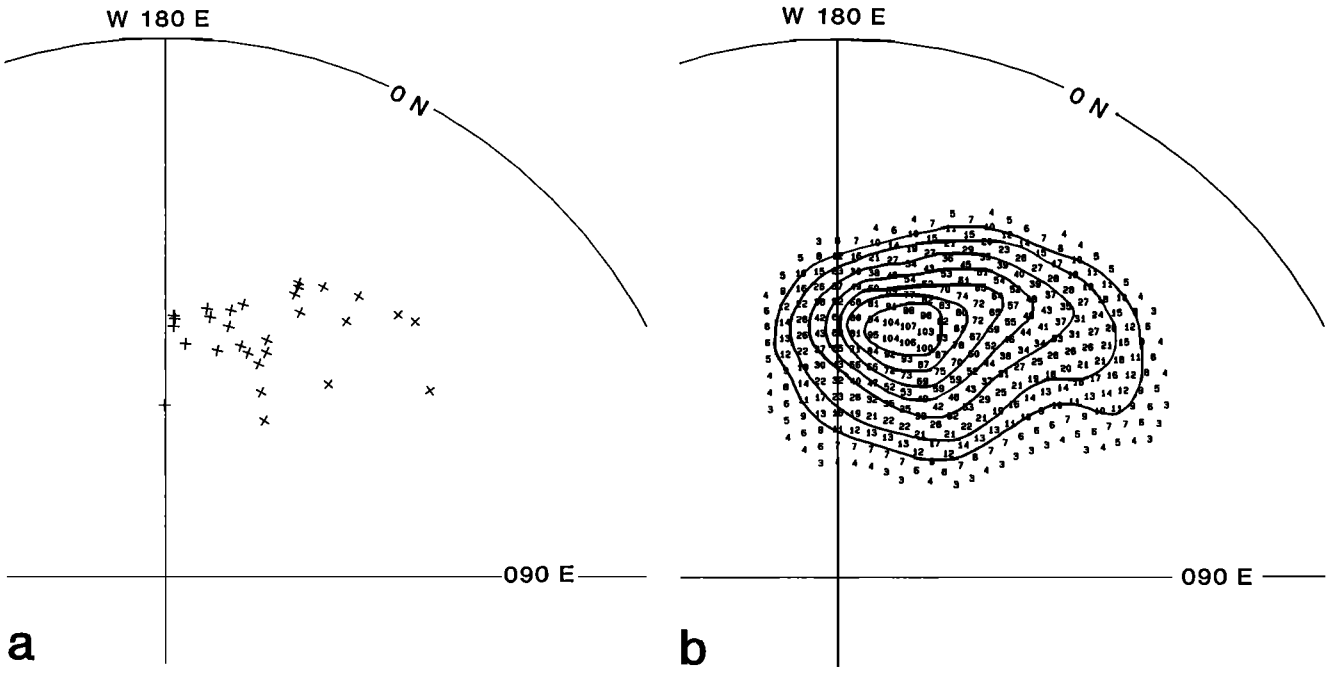


Fig. 7. The third, reversed component of the Rote Fäule (MM3) in 29 samples (70 specimens) represented as (*a*) point plots and (*b*) contoured density diagrams of the north virtual geomagnetic poles of the measured moments. Contours in Figure 7*b* are E+2σ, E+4σ, E+8σ...E+24σ (see Figure 5 caption).

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TABLE 3. List of Paleomagnetic North Poles Used in Calculating the APW Paths for Europe and Russia

Ident.	Latitude	Longitude	Wt.	Age
Carboniferous				
0- 0	38°N	167°E	3	287*
6- 185	46°N	152°E	1	342
6- 158	47°N	160°E	1	342
6- 159	41°N	158°E	1	342
6- 160	46°N	152°E	1	342
6- 184	49°N	164°E	1	342
6- 182	45°N	156°E	1	342
6- 250	43°N	159°E	1	327
6- 183	41°N	158°E	1	342
6- 148	33°N	158°E	1	302
6- 149	34°N	170°E	1	302
6- 186	33°N	158°E	1	302
6- 142	42°N	168°E	1	300
6- 176	38°N	176°E	1	289
6- 218	42°N	167°E	1	289
6- 150	29°N	155°E	1	303
6- 151	31°N	168°E	1	303
6- 187	20°N	150°E	1	303
6- 222	32°N	166°E	1	303
6- 188	23°N	179°E	1	315
6- 219	31°N	177°W	1	315
6- 154	28°N	177°E	1	315
6- 251	24°N	173°E	1	319
6- 287	42°N	150°E	1	300
6- 334	31°N	166°E	1	300
6- 335	17°N	168°E	1	300
6- 336	29°N	173°E	1	315
6- 337	19°N	166°E	1	315
6- 338	29°N	177°W	1	315
6- 339	32°N	177°E	1	315
6- 152	33°N	170°W	1	300
6- 153	24°N	166°E	1	300
6- 341	36°N	165°W	1	300
6- 342	35°N	165°W	1	300
6- 343	28°N	169°W	1	315
6- 344	28°N	152°W	1	315
6- 143	40°N	177°W	1	300
6- 179	40°N	178°W	1	300
6- 144	25°N	174°W	1	300
6- 181	38°N	170°E	1	289
6- 239	44°N	171°E	1	300
6- 206	38°N	178°E	2	320
6- 205	21°N	162°W	1	340
6- 331	27°N	167°E	2	358
6- 329	48°N	173°W	2	323
6- 330	42°N	181°E	2	342
6- 175	18°N	161°E	3	340
6- 104	39°N	161°E	3	297*
6- 311	45°N	150°E	3	286
6- 126	31°N	174°E	1	287*
6- 127	38°N	166°E	1	279*
6- 128	37°N	174°E	1	280
6- 131	36°N	174°E	3	286
6- 324	39°N	169°E	3	286
6- 95	31°N	152°E	1	300
6- 97	36°N	154°E	1	296
6- 98	39°N	177°E	1	300
6- 99	40°N	154°E	1	288
6- 100	40°N	169°E	1	288
6- 101	45°N	174°E	1	288
6- 94	43°N	174°E	3	290
6- 89	40°N	156°E	1	286
Permian				
0- 0	52°N	142°E	3	254
0- 0	48°N	151°E	3	254
0- 0	52°N	144°E	3	254

TABLE 3. (continued)

Ident.	Latitude	Longitude	Wt.	Age
Permian (continued)				
0- 0	46°N	155°E	3	263
0- 0	41°N	169°E	3	280*
0- 0	46°N	164°E	3	273
0- 0	44°N	172°E	3	277
0- 0	31°N	166°E	3	278
7- 14	43°N	164°E	1	280
7- 88	50°N	149°E	1	280
7- 87	46°N	165°E	1	280
7- 89	46°N	165°E	1	260
7- 16	36°N	175°E	1	280
7- 17	37°N	167°E	1	280
7- 18	34°N	163°E	1	280
7- 137	46°N	166°E	2	260
7- 13	47°N	157°E	3	280
7- 295	54°N	172°E	3	253
7- 103	43°N	165°E	3	280
7- 104	40°N	156°E	1	286
7- 113	43°N	175°E	3	280
7- 1	46°N	142°E	1	254
7- 2	45°N	131°E	1	254
7- 3	53°N	165°E	1	254
7- 6	38°N	162°E	1	260
7- 7	43°N	168°E	1	280
7- 54	41°N	169°E	1	280
7- 274	37°N	165°E	1	280
7- 164	47°N	148°E	3	253
7- 8	45°N	175°W	1	268
7- 9	40°N	170°W	2	275
7- 10	40°N	168°E	1	280
7- 11	46°N	167°E	2	280
7- 216	48°N	174°E	3	280
7- 217	31°N	154°E	3	268
7- 69	36°N	150°E	2	268
7- 122	40°N	158°E	1	275
7- 123	41°N	162°E	1	275
7- 124	40°N	168°E	1	275
7- 125	41°N	159°E	1	275
7- 126	42°N	169°E	1	266
7- 127	38°N	170°E	1	275
7- 129	38°N	178°E	2	280
7- 134	38°N	149°E	3	280
7- 158	31°N	159°E	2	286
7- 159	30°N	160°E	2	280
7- 235	40°N	162°E	2	281
7- 236	40°N	166°E	3	283
7- 252	37°N	162°E	3	272
7- 146	39°N	159°E	1	253
7- 227	42°N	164°E	2	255
7- 221	46°N	170°E	1	251
7- 222	44°N	164°E	1	249
7- 24	48°N	164°E	1	249
7- 186	49°N	167°E	1	249
7- 188	45°N	173°E	1	249
7- 259	48°N	165°E	1	249
7- 261	47°N	161°E	1	249
7- 257	48°N	161°E	1	249
7- 262	50°N	160°E	1	249
7- 263	50°N	160°E	1	249
7- 26	45°N	171°E	1	252
7- 27	41°N	172°E	1	252
7- 28	45°N	167°E	1	252
7- 190	39°N	167°E	1	252
7- 239	43°N	171°E	1	252
7- 256	43°N	161°E	1	252
7- 229	42°N	168°E	3	253
7- 238	40°N	160°E	1	251
7- 157	34°N	170°E	1	251
7- 264	44°N	167°E	1	251
7- 135	43°N	167°E	1	253
7- 32	46°N	173°E	1	255
7- 33	40°N	167°E	1	255

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TABLE 3. (continued)

Ident.	Latitude	Longitude	Wt.	Age
Permian (continued)				
7- 240	43°N	168°E	1	255
7- 243	56°N	165°E	1	254
7- 244	55°N	173°E	1	254
7- 245	50°N	169°E	1	256
7- 230	41°N	173°E	1	255
7- 156	32°N	172°E	1	255
7- 232	39°N	168°E	1	253
7- 233	37°N	168°E	1	253
7- 247	45°N	180°E	1	253
Triassic				
0- 0	49°N	146°E	3	244
0- 0	43°N	134°E	3	247
0- 0	59°N	146°E	3	247
8- 6	43°N	131°E	3	225
8- 7	44°N	134°E	1	225
8- 123	56°N	148°E	1	222
8- 3	44°N	151°E	2	245
8- 4	55°N	159°E	2	245
8- 116	46°N	155°E	1	235
8- 124	51°N	170°W	2	248
8- 187	57°N	144°E	3	245
8- 56	51°N	146°E	1	245
8- 207	60°N	157°E	2	235
8- 189	51°N	151°E	3	245
8- 12	49°N	158°E	1	245
8- 141	43°N	155°E	1	245
8- 193	41°N	149°E	1	245
8- 155	50°N	174°E	1	245
8- 190	52°N	167°E	1	245
8- 195	49°N	159°E	1	245
8- 13	54°N	164°E	1	246
8- 139	45°N	152°E	1	245
8- 140	51°N	164°E	1	245
8- 192	45°N	157°E	1	245
8- 221	53°N	158°E	1	247
8- 224	52°N	174°E	1	245
8- 225	49°N	168°E	1	245
8- 145	35°N	158°E	1	231
8- 181	44°N	146°E	2	222
8- 223	57°N	162°E	1	244
Jurassic				
0- 0	44°N	110°E	3	162*
0- 0	70°N	131°E	3	180*
9- 1	70°N	133°W	1	188
9- 2	71°N	111°E	1	189
9- 164	61°N	150°W	2	153
9- 85	63°N	114°E	3	213
9- 89	65°N	144°E	3	200
9- 107	64°N	138°E	2	163
9- 129	72°N	113°W	2	175
9- 122	81°N	169°E	2	153
9- 125	63°N	117°E	2	172
Cretaceous				
10- 1	79°N	115°W	2	129
10- 2	87°N	151°E	1	128
10- 3	86°N	39°E	1	128
10- 33	84°N	11°E	1	128
10- 213	75°N	125°W	3	81
10- 214	77°N	107°E	2	81
10- 26	87°N	13°W	2	81
10- 88	79°N	173°W	3	65
10- 51	63°N	104°W	3	131
10- 250	85°N	52°E	3	79
10- 56	66°N	177°E	1	77
10- 143	66°N	56°W	2	97

TABLE 3. (continued)

Ident.	Latitude	Longitude	Wt.	Age
Cretaceous (continued)				
10- 135	80°N	21°E	2	116
10- 145	74°N	140°W	2	127
10- 144	77°N	153°W	3	114
10- 162	77°N	175°W	2	88
10- 264	67°N	75°E	1	75
10- 265	75°N	188°W	1	81
10- 266	73°N	129°E	1	85
10- 267	62°N	116°W	1	90
10- 105	52°N	158°W	2	77
10- 278	60°N	141°E	2	105
10- 100	60°N	167°E	2	110
10- 179	81°N	166°E	2	111
Tertiary				
11- 18	73°N	135°E	1	50*
11- 19	80°N	158°W	1	50*
11- 22	79°N	158°E	3	50*
11- 23	76°N	138°E	2	50*
11- 199	70°N	163°E	3	50*
11- 535	71°N	126°E	3	59*
11- 27	75°N	130°E	1	50
11- 156	73°N	163°W	1	46*
11- 17	78°N	149°E	3	60
11- 16	72°N	133°E	1	60
11- 532	75°N	59°W	1	52
11- 533	73°N	174°E	1	52
11- 564	75°N	62°W	3	59*
11- 13	69°N	165°E	2	55
11- 14	69°N	171°E	1	55
11- 12	74°N	157°E	1	55
11- 506	73°N	165°E	3	46
11- 237	77°N	149°W	3	34
11- 580	62°N	142°E	2	33
11- 675	79°N	147°E	3	45
11- 656	68°N	165°E	2	46
11- 667	71°N	158°E	3	21*
11- 693	80°N	143°E	3	48*
11- 326	79°N	121°W	3	32*
11- 695	61°N	122°E	3	45
11- 335	64°N	146°E	3	34
11- 703	65°N	148°E	2	45
11- 635	67°N	126°W	2	45
11- 630	60°N	158°E	2	44
11- 631	66°N	134°W	1	42
11- 636	75°N	156°W	1	63
11- 637	77°N	114°W	1	60
11- 655	82°N	106°E	2	2*
11- 221	77°N	98°E	2	25
11- 194	79°N	140°E	1	2
11- 195	83°N	172°E	1	3
11- 196	84°N	176°W	1	4
11- 197	85°N	109°E	1	8
11- 385	83°N	133°E	2	9*
11- 386	58°N	131°E	1	5
11- 387	62°N	113°E	2	5
11- 6	76°N	164°W	1	6
11- 7	82°N	142°E	1	12
11- 10	86°N	168°E	1	12
11- 11	76°N	160°E	2	12
11- 110	73°N	92°E	1	12
11- 111	68°N	132°E	1	12
11- 112	65°N	104°E	1	12
11- 113	70°N	108°E	3	12
11- 116	87°N	141°W	2	12
11- 114	81°N	151°E	1	12
11- 115	83°N	163°E	3	12
11- 228	70°N	171°E	1	17*
11- 208	62°N	174°E	1	6
11- 300	71°N	171°E	1	6*
11- 692	84°N	177°E	3	8*

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TABLE 3. (continued)

Ident.	Latitude	Longitude	Wt.	Age
Tertiary (continued)				
11- 399	83°N	163°W	3	7
11- 322	72°N	161°W	2	8
11- 153	78°N	143°E	3	15*
11- 220	75°N	123°E	3	25
11- 253	82°N	176°W	3	15
11- 299	67°N	36°E	3	12
11- 236	79°N	54°E	3	15
11- 712	80°N	67°W	2	12
11- 264	86°N	157°E	3	21
11- 265	80°N	130°E	3	5
11- 144	85°N	164°W	2	25
11- 389	77°N	35°E	1	14
11- 713	67°N	8°E	3	3
11- 219	84°N	122°E	1	8
11- 218	81°N	142°E	1	2
11- 210	87°N	163°W	3	8
11- 262	78°N	159°W	3	11
11- 456	84°N	176°W	2	15
11- 455	73°N	166°W	3	15
11- 324	83°N	70°W	3	8
11- 261	85°N	61°W	3	8
11- 308	77°N	70°E	3	3
11- 489	78°N	180°E	3	13
11- 545	81°N	108°E	2	3
11- 479	85°N	141°W	3	10*
11- 613	82°N	45°W	2	3
11- 462	86°N	15°W	2	3
11- 465	68°N	12°E	2	3
11- 623	82°N	85°E	3	3
11- 466	77°N	33°E	2	12
11- 467	83°N	30°W	2	3
11- 550	71°N	173°E	2	3
11- 469	80°N	150°W	2	3
11- 342	74°N	174°W	1	15
11- 373	77°N	159°W	1	3
11- 628	76°N	122°W	3	12
11- 629	71°N	164°E	1	3
11- 470	78°N	153°W	3	2
11- 715	68°N	168°E	3	3
11- 724	76°N	150°W	1	3
11- 732	79°N	169°W	3	3
Quaternary				
12- 6	87°N	180°E	1	0
12- 7	77°N	170°E	1	0
12- 8	82°N	172°E	2	0
12- 1	78°N	93°E	1	1
12- 2	72°N	84°E	1	1
12- 3	85°N	106°W	1	1
12- 97	83°N	135°W	1	1
12- 144	86°N	148°W	3	1
12- 145	78°N	173°W	3	1
12- 138	80°N	132°E	3	1
12- 152	81°N	156°W	3	1
12- 9	86°N	126°E	1	1
12- 257	89°N	119°W	3	0
12- 10	84°N	141°W	2	0
12- 11	82°N	160°W	1	0
12- 12	86°N	142°W	1	0
12- 13	89°N	156°E	1	0
12- 14	80°N	49°E	1	0
12- 4	83°N	172°W	2	0
12- 100	81°N	142°E	3	1
12- 106	79°N	50°E	3	1
12- 105	82°N	145°E	3	1
12- 186	80°N	37°E	3	1
12- 217	82°N	1°E	2	1
12- 228	82°N	113°E	2	2
12- 184	79°N	149°E	2	1
12- 19	75°N	159°W	1	1
12- 255	70°N	173°W	2	1
12- 219	78°N	80°W	2	1

TABLE 3. (continued)

Ident.	Latitude	Longitude	Wt.	Age
Quaternary (continued)				
12- 220	87°N	135°W	2	1
12- 229	79°N	152°W	3	2
12- 218	89°N	101°W	3	2
12- 279	83°N	158°W	2	1

Identification numbers (Ident.) are those of *Hicken et al.* [1972] and *Irving et al.* [1976*a*, *b*, *c*]. Poles with identification numbers of 0- 0 are taken from *Morel and Irving* [1981]. Weighting (Wt.), given as 3, 2, or 1, corresponds to the reliability (A**, A*, or A, B) assigned by *Irving et al.* [1976*a*, *b*, *c*]. Ages in Ma are recalculated from the time scale of *Harland et al.* [1964] to that of *Harland et al.* [1982].
*Radiometric ages are used as is.

ferentiation can be obtained with the McFadden-Lowes test.) This loop of the polar path coincides with the peak of northerly thrusting and folding in the Hercynian orogeny, which lasted from about 340 to 300 Ma [*Ziegler*, 1982].

After Lower Rotliegendes sedimentation (around 290-270 Ma), during which the paleopoles were stationary at about 40°N, 166°E, northward polar wander recurred during Upper Rotliegendes and Zechstein sedimentation to 47°N, 162°E at 250 Ma, the end of the Permian. One very noteworthy feature of the path is the large spread between the 240- and 220-Ma poles, implying rapid westerly polar movement during the Triassic from 50°N, 159°E at 240 Ma to 51°N, 136°E at 220 Ma (more than 15° in 20 Ma).

The poles moved rapidly during the Early Jurassic to 74°N, 149°E at 190 Ma, followed by a poorly defined easterly motion to 72°N, 200°E at 150 Ma, the Jurassic-Cretaceous boundary. The apparent movement is so rapid and the sampling so poor in the Jurassic that the APW path in this time period is not reliable. From the Early Cretaceous (130 Ma) through the Cenozoic (Table 4), the APW path loops around several times but always within 16° of the geographic north pole, and movements of the continents themselves are not discernible.

The 30-Ma Interval Path

Comparison of the 30-Ma path (Figures 8, 9, and Table 4) with the 20-Ma path shows that the 30-Ma path records all the major loops and rapid motion segments present in the more detailed 20-Ma path, but because of the wider sampling interval, it gives a more generalized estimate of the path. The rapid westerly Triassic movement begins slightly later at 230 Ma instead of at 240 Ma, and the northerly Jurassic movement begins at 210 Ma instead of 220 Ma. The 30-Ma path is useful in providing a check on the reliability of events shown in the 20-Ma path, which is more susceptible to both real and imaginary excursions.

The continental reconstructions of *Kanasewich et al.* [1978], *Scotese et al.* [1979], and *Smith* [1981] suggest that the apparent polar wander along the paths, northwesterly between 320 Ma (Carboniferous) and 250 Ma (latest Permian) and westerly between 250 and 210 Ma (Triassic-Jurassic boundary) (Figure 9) is the result of actual translation of the Europe and Russia continental plate, with little rotation.

Both APW paths indicate rapid polar movement during the Permian and Triassic, which is the period of interest in this study, and the good data set of catalog poles constrains the paths well, presenting an ideal situation for dating diagenetic events with confidence.

MAGNETIC COMPONENTS AND THE APW PATHS

Age and Origin of the VS1 and VS2 Europe Components

The VS1 component lies off the APW path, and as the lowest temperature component isolated, it represents a highly unstable

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TABLE 4. Apparent Polar Wander Paths of Europe and Russia From 350 Ma to the Present, using 20-Ma and 30-Ma Overlapping Time Intervals

	20-Ma APW Path						30-Ma APW Path					
Ma	Latitude	Longitude	N	k	A_{63}, deg	A_{95}, deg	Latitude	Longitude	N	k	A_{63}, deg	A_{95}, deg
350	39.3°N	165.0°E	11	27	4.7	8.1	39.3°N	165.0°E	11	27	4.7	8.1
340	40.6°N	164.8°E	10	26	5.0	8.7	40.8°N	164.2°E	11	29	4.6	7.9
330	34.6°N	176.7°E	5	17	8.7	15.0	35.5°N	173.6°E	23	26	3.3	5.7
320	31.3°N	179.5°E	13	39	3.6	6.2	31.3°N	179.5°E	13	39	3.6	6.2
310	32.0°N	173.7°E	32	35	2.4	4.2	32.8°N	173.1°E	35	34	2.3	4.0
300	34.5°N	169.1°E	24	37	2.7	4.7	34.7°N	170.6°E	47	39	1.9	3.3
290	39.0°N	168.1°E	53	61	1.4	2.5	38.2°N	167.2°E	69	62	1.2	2.1
280	40.3°N	166.2°E	48	117	1.1	1.9	40.2°N	166.0°E	52	104	1.1	1.9
270	40.9°N	165.7°E	39	100	1.3	2.2	41.1°N	166.2°E	48	111	1.1	1.9
260	44.3°N	163.8°E	41	71	1.5	2.6	46.0°N	162.7°E	79	72	1.1	1.9
250	47.2°N	162.0°E	68	77	1.1	1.9	47.4°N	161.8°E	71	78	1.1	1.9
240	49.6°N	159.1°E	35	92	1.4	2.5	47.3°N	160.7°E	69	66	1.2	2.1
230	47.3°N	146.7°E	7	51	4.3	7.4	49.2°N	154.3°E	25	79	1.8	3.2
220	50.5°N	135.7°E	5	46	5.4	9.3	49.7°N	143.9°E	8	36	4.8	8.2
210	64.8°N	128.5°E	2	76	6.6	11.4	53.0°N	136.7°E	6	43	5.0	8.7
200	65.0°N	144.0°E	1	1	81.0	140.0	72.9°N	140.7°E	4	20	9.0	15.5
190	73.9°N	148.6°E	4	26	8.0	13.8	77.1°N	165.1°E	5	20	8.0	13.9
180	77.9°N	154.4°E	5	18	8.6	14.9	77.9°N	154.4°E	5	18	8.6	14.9
170	68.2°N	129.8°E	5	14	9.7	16.8	68.2°N	129.8°E	5	14	9.7	16.8
160	68.3°N	146.5°E	4	11	12.4	21.5	72.1°N	149.2°E	6	12	9.7	16.8
150	71.8°N	159.7°W	2	25	11.4	19.8	68.3°N	146.5°E	4	11	12.4	21.5
140	63.0°N	104.0°W	1	1	81.0	140.0	80.7°N	133.4°W	8	33	5.0	8.6
130	82.6°N	112.5°W	6	40	5.3	9.1	84.5°N	101.4°W	7	38	5.0	8.6
120	84.3°N	170.1°W	9	38	4.4	7.6	82.5°N	168.5°W	11	25	5.0	8.5
110	76.2°N	158.3°E	5	22	7.7	13.4	81.4°N	174.0°E	6	15	8.7	15.0
100	75.4°N	158.8°W	4	8	13.9	24.1	77.6°N	173.3°W	8	17	6.9	11.9
90	82.8°N	147.0°W	8	23	6.0	10.4	80.5°N	179.2°E	13	17	5.5	9.5
80	79.8°N	178.5°E	11	20	5.5	9.4	79.8°N	179.3°E	12	22	5.0	8.6
70	78.4°N	172.2°E	9	23	5.7	9.8	79.2°N	166.9°E	19	30	3.4	5.9
60	78.9°N	166.8°E	18	47	2.8	4.8	77.8°N	161.5°E	27	39	2.5	4.3
50	76.6°N	166.1°E	26	38	2.6	4.5	76.8°N	168.2°E	28	40	2.4	4.2
40	74.4°N	164.1°E	20	42	2.8	4.9	75.8°N	163.4°E	28	41	2.4	4.1
30	78.0°N	153.5°E	9	42	4.2	7.2	76.2°N	164.9°E	24	37	2.7	4.7
20	82.0°N	146.0°E	32	42	2.2	3.8	81.5°N	150.3°E	49	40	1.8	3.1
10	84.1°N	153.0°E	84	45	1.3	2.3	83.9°N	151.6°E	89	46	1.3	2.2
0	85.0°N	157.4°E	58	50	1.5	2.6	84.7°N	150.0°E	81	47	1.3	2.3

All 340 poles used are unit weighted. N, k, A_{63}, and A_{95} are as in Table 1. Sources are *Hicken et al.* [1972], *Irving et al.* [1976*a*, *b*, *c*], and *Morel and Irving* [1981].

component. Assuming that it is highly viscous, it likely reflects a near-contemporaneous magnetic field direction.

The VS2 component is very similar (even at 50% confidence, the null hypothesis that they are the same cannot be rejected) to 180-, 110-, 20-, and 10-Ma directions on both 20-Ma and 30-Ma APW paths; to 100 Ma on the 20-Ma path; and 190 Ma on the 30-Ma path. Therefore the component can be either Jurassic (180 Ma), Cretaceous (110 Ma), or Miocene (20–10 Ma) in age with equal probability. The 180- and 110-Ma directions range up to 6.4° away from the VS2 component, and are statistically similar to VS2 only because they are so poorly constrained (A_{63} of 7°–9°). Because the VS2 component could be 10°–12° lower in latitude and still be statistically similar to the 180- and 110-Ma ages, the 20- and 10-Ma directions, being much better constrained, are considered to be more defensible estimates.

This second component might possibly be a viscous partial thermal remanent magnetization (VPTRM) [*Chamalaun*, 1964] carried by magnetite and formed upon uplift and cooling [*Pullaiah et al.*, 1975] during the Cretaceous and later. The Zechstein rocks would have cooled from a maximum temperature of 110°–140°C [*Gondek*, 1980; *Majorowicz et al.*, 1983]. The VPTRM should include blocking temperatures well above this range [*Pullaiah et al.*, 1975; *Dunlop*, 1983], possibly as high as 350°C. However, there is no indication in these rocks of a magnetic remanence component in magnetite above 350°C as one might expect if there were significant magnetite present.

Another possible explanation is that VS2 is a CRM carried by goethite or maghemite. *Strangway et al.* [1968, 1969] showed that goethite can retain a remanent magnetization at temperatures up to 250°–350°C and does not lose all its magnetization until 400°–450°C, even though it should retain no magnetization above its Neel temperature of about 120°C (Hedley, 1968). The pyrite-rich basal Zechstein should have formed hydrous oxides of iron, like goethite, upon uplift and oxidation after the Cretaceous, and the CRM thus produced would reflect the ambient field to demagnetization temperatures of 350°–450°C. (Outcrop samples often have a yellowish weathering rind on them, and locally contain orange spots and patches.) Maghemite can be formed by low-temperature oxidation of magnetite in marine sediments [*Stacey and Banerjee*, 1974] and it also converts to hematite over a range of temperatures up to 350°C. Since the VS2 component is stable to 350°–450°C and best corresponds to a Miocene age, it is likely a CRM formed by a mixture of hydroxides from weathering. *Vaughan and Turner* [1980] concluded that the remanence in the Marl Slate (Kupferschiefer equivalent) from a quarry in England reflected the present magnetic field and was caused by goethite.

Age and Origin of the Rote Fäule Component From Polish Mines

The reversed MM3 remanence component of the RF fits the 240-Ma position extremely well on the 20-Ma path and the 230-

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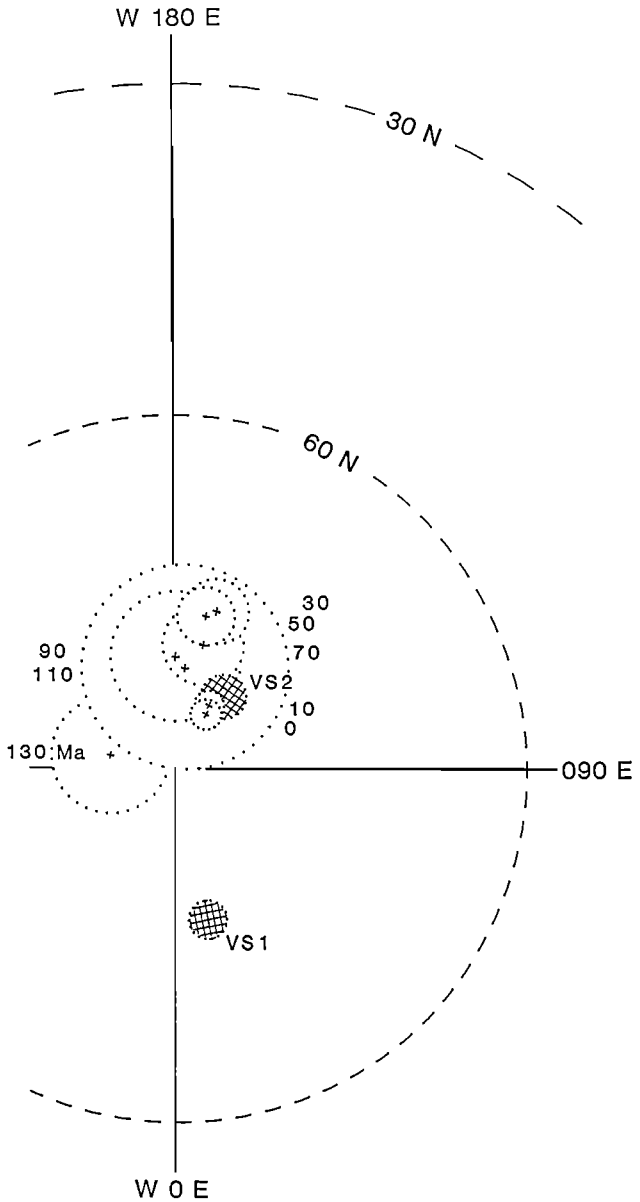


Fig. 8. The 30-Ma apparent polar wander path of Europe and Russia described by running averages of unit-weighted published paleopoles, with their A_{63} circles, for 130–0 Ma, using a 30-Ma overlapping time interval and the time scale of *Harland et al.* [1982]. To avoid clutter, poles are plotted only every 20 Ma: Table 4 has complete listing. The first and second magnetic components (VS1 and VS2) of the pyritic basal Zechstein are shown. Sources are *Hicken et al.* [1972], *Irving et al.* [1976*a*, *b*, *c*], and *Morel and Irving* [1981].

Ma position on the 30-Ma path (the null hypothesis that they are the same cannot be rejected at 50% to 99% confidence levels) (Figure 9). At the 95% confidence level, the MM3 component is also the same as the 250- and 230-Ma positions on the 20-Ma path, and as the 250-, 240-, and 220-Ma positions on the 30-Ma path. Conversely, this component is different (can be rejected with 95% confidence) from the 220- and 260-Ma positions on the 20-Ma path, and from the 210- and 260-Ma positions on the 30-Ma path. It is not statistically possible, using the usual 95% confidence level, to correlate the stable end component of the Rote Fäule with a basal Zechstein age of 260 Ma. The best estimate of the mean age of the stable magnetic component of the RF is Middle Triassic, between 240 Ma (Late Anisian) and 230 Ma (Early Carnian) using the time scale of *Harland et al.* [1982].

On earlier versions of the APW paths, constructed using the time scale of *Harland et al.* [1964], the RF component was coincident with the 210-Ma direction, also Middle Triassic, on both 20-Ma and 30-Ma paths.

The Triassic tectonic movements in the North Sudetic Basin were vertical, and tilting and folding occurred later during the Alpine orogeny [*Milewicz*, 1968; *Lisiakiewicz*, 1969]. Because the RF is displaced by Alpine tectonism, it can be assumed that the RF formed before folding, and the structural corrections made on a few samples are valid. The MM3 component is isolated typically above 500°C and persists up to the Curie temperature of hematite (680°C), indicating that this component is the primary remanence in the RF zones. The Triassic age suggests that the MM3 is a secondary CRM component acquired during the RF hematite and copper mineralization event, and that the mineralization is Triassic in age.

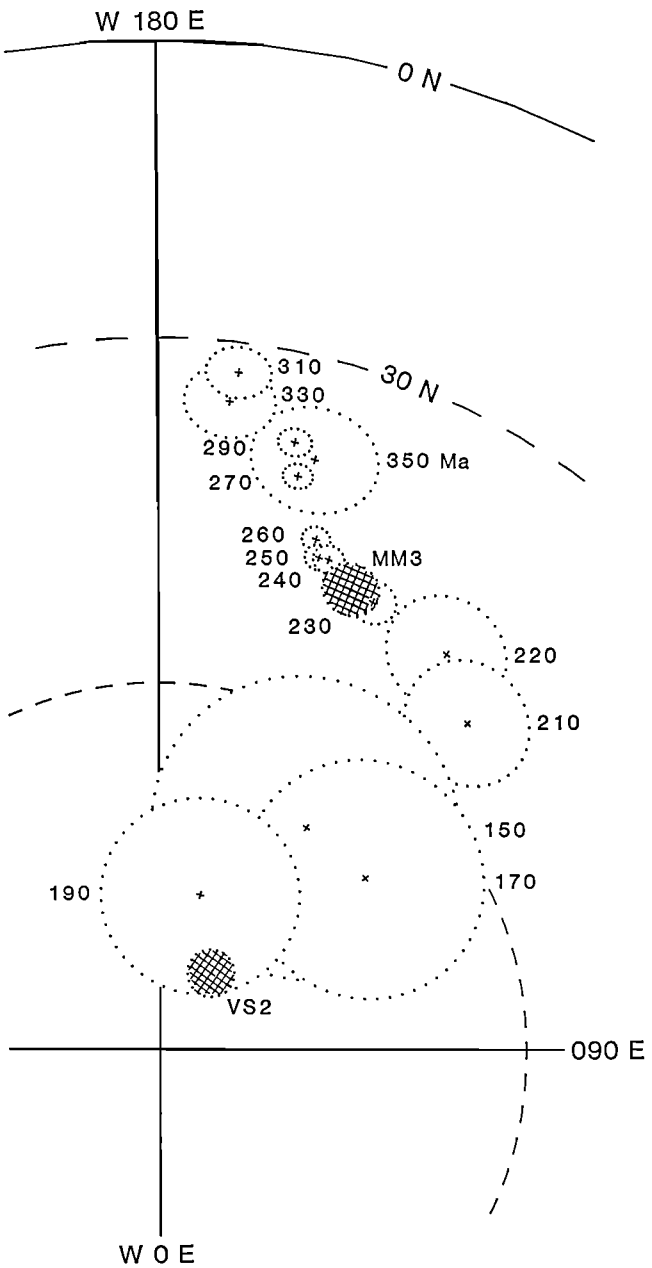


Fig. 9 The 30-Ma APW path for 350–150 Ma, and the stable end component (MM3) of the Rote Fäule zones of the Kupferschiefer ore deposits (see Figure 8 caption).

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The trend toward reversed directions in certain of the Lubin and Polkowice copper ore specimens could be related to the mineralizing event because this behavior does not occur in the barren pyritic rocks in Germany and England, and because it is more evident near the base of the Kupferschiefer closer to the RF zones.

No negative inclinations in the unweathered sulphide zones in the mines were found in this study, and the trend to reversed directions occurred only at elevated temperatures. No evidence of the primary detrital remanent magnetization and rapid changes in the Permian geomagnetic field suggested by *Turner and Vaughan* [1977] and *Vaughan and Turner* [1980] was not found in our study.

At first glance, the reversed polarity of the RF component appears to be inconsistent with the interval of mixed polarities extending from the latest Permian [*McElhinny*, 1973] or Early Triassic [*McMahon and Strangway*, 1968] through to the Early Cretaceous [*Irving and Pullaiah*, 1976; *Harland et al.*, 1982]. However, a look at the catalogs of poles [*Irving et al.*, 1976*b*] shows that Triassic rocks can contain either all normal, all reversed, or mixed polarities. Therefore the reversed direction of the third component neither supports nor refutes a Triassic age for the RF.

ORIGIN OF THE ROTE FÄULE AND ORE

The ore textures and geologic controls noted in the introduction and the Middle Triassic paleomagnetic age suggest that the RF-ore systems were formed during late diagenesis by brines channelled up along the buried basement highs to the overlying Zechstein, carrying metals leached from the Rotliegendes volcanic detritus, as suggested by *Jowett* [1983] and *Jowett et al.* [1986]. The fluids oxidized the original pyritic rock to form the RF, and precipitated base and precious metals on the reduced side of the redox front. *Salski* [1977] also concluded that the dilatant sulphide veinlets in the Lubin district were formed during Middle to Late Triassic (Early Kimmerian) tectonism, although he considers these to be remobilizations of the otherwise syngenetic ore (W. Salski, personal communication, 1983). This Middle Triassic age supports the epigenetic timing suggested by *Lisiakiewicz* [1969] for U-Cu-Pb-Zn mineralization in the Kupferschiefer. He concluded that they were formed by hydrothermal solutions from both magmatic and formational sources, which migrated in the Late Triassic during the early stages of the Kimmerian event.

A flow-through model, as suggested by *White* [1971] for the White Pine (Michigan) deposit, with the fluids migrating vertically through the reduced beds is precluded by the presence of thick beds of impermeable anhydrite and salt in the Lower Zechstein. The metal zoning suggests a model of convection instead, with the fluids turning along the base of the Zechstein and back down into the Rotliegendes toward the basin center. The convection model proposed by *Jowett* [1986] relied on a regional thermal anomaly at the base of the lithosphere, and differential thermal conductivities between the basement highs and shaley centers of the basin, to create a lateral temperature gradient, which initiated unicellular convection. Triassic continental rifting in Poland during the opening of the Tethys ocean to the south [*Ziegler*, 1982], with the accompanying thinning of the lithosphere and rising of the asthenosphere, may have been the source of this thermal anomaly.

CONCLUSIONS

Apparent polar wander paths constructed from published catalogs show that northern Europe migrated consistently northward from the Carboniferous through to the Cretaceous. Movement from the Carboniferous to the Triassic-Jurassic boundary is considered to be lateral continental drift or translation, with little rotation. Control of the paleopole positions is good for the Permian and Triassic, but less so during the Jurassic and Cretaceous due to lack of suitable rocks and to rapid apparent polar motion.

Two normal magnetic components are present in the sulphide ore and barren pyritic rocks of the basal Zechstein in Europe, but no recoverable original remanence remains. The first component removed during the initial demagnetization step represents an unstable, likely highly viscous, component which lies off the APW path. The second component, revealed up to 350°–400°C, is likely Miocene (20–10 Ma) in age and carried by goethite or other hydroxide produced during Alpine age uplift and weathering.

A third, reversed component, revealed above 500°C in the hematitic RF zones of the ore deposits, occurs as a stable end point and may possibly occur in the copper zone as an unrecoverable remanence. Its direction corresponds with Middle to Late Triassic (240–230 Ma) ages on the APW paths.

Geologic controls and the Triassic paleomagnetic age of mineralization support a model of late diagenetic convection of metalliferous fluids within the Rotliegendes to form the Kupferschiefer ore deposits. The coincidence between the timing of the ore and continental rifting suggests that thinning of the lithosphere was a likely source of the thermal anomaly necessary to initiate convection.

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